

Twin Support Vector Machine (TSVM) Overview

Introduction

The Twin Support Vector Machine (TSVM) is a variation of the traditional Support Vector Machine (SVM) that introduces two non-parallel hyperplanes to enhance classification performance, particularly in scenarios with overlapping classes.

TSVM Formulation

Given a dataset consisting of two classes, let $\mathcal{D}_1$ denote the positive class (where $y_i = +1$) and $\mathcal{D}_{-1}$ denote the negative class (where $y_i = -1$). The TSVM formulation is given as:

$$\begin{aligned} \min_{b_k, c_k, \xi_k} \quad & \sum_{i \in \mathcal{D}_k} (b_k^T x_i + c_k)^2 + C_k \sum_{i \in \mathcal{D}_{-k}} \xi_{k_i} \\ \text{subject to} \quad & 1 - y_i (b_k^T x_i + c_k) \leq \xi_{k_i}, \quad \forall i \in \mathcal{D}_{-k}, \\ & b_k \in \mathbb{R}^n, \quad c_k \in \mathbb{R}, \quad \xi_k \in \mathbb{R}_+^{|\mathcal{D}_{-k}|}. \end{aligned}$$

Where: - n is the number of features (dimensions) of the input data. - $|\mathcal{D}_{-k}|$ is the number of samples in the negative class.

Standard Formulation for CVXPY

To express the TSVM in a standard form suitable for optimization frameworks like CVXPY, we rewrite it as:

$$\begin{aligned} \min \quad & \frac{1}{2} x^T P x + q^T x \\ \text{subject to} \quad & Gx \leq h \end{aligned}$$

Definitions

- **Quadratic Coefficients P **:

$$P = \begin{bmatrix} 2X_k^T X_k & 2X_k^T \mathbf{1}_{m_k} \\ 2\mathbf{1}_{m_k}^T X_k & 0 \end{bmatrix}$$

Where: - $P \in \mathbb{R}^{(n+1+|\mathcal{D}_{-k}|) \times (n+1+|\mathcal{D}_{-k}|)}$

- **Linear Coefficients q^{**} :

$$q = \begin{bmatrix} 0 \\ 0 \\ C_{-k} \mathbf{1}_{|\mathcal{D}_{-k}|} \end{bmatrix}$$

Where: - $q \in \mathbb{R}^{(n+1+|\mathcal{D}_{-k}|)}$

- **Inequality Constraints G^{**} :

$$G = \begin{bmatrix} -y_1 X_{-k}^T & -y_1 & -1 \\ -y_2 X_{-k}^T & -y_2 & -1 \\ \vdots & \vdots & \vdots \\ -y_{|\mathcal{D}_{-k}|} X_{-k}^T & -y_{|\mathcal{D}_{-k}|} & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \\ \vdots & \vdots & \vdots \end{bmatrix}$$

Where: - The number of rows in G is $|\mathcal{D}_{-k}| + |\mathcal{D}_{-k}|$ (one for each sample in the negative class plus one for each slack variable).

- **Right-Hand Side Vector h^{**} :

$$h = \begin{bmatrix} -1 \\ -1 \\ \vdots \\ -1 \\ 0 \\ 0 \\ \vdots \end{bmatrix} \in \mathbb{R}^{|\mathcal{D}_{-k}| + |\mathcal{D}_{-k}|}$$

Conclusion

The Twin Support Vector Machine (TSVM) formulation can be expressed as a standard quadratic program, making it suitable for optimization libraries such as CVXPY. By defining the matrices P , q , G , and vector h with clear sizes and relationships, TSVM can be solved efficiently while accommodating the challenges presented by overlapping classes.